5.02 Discrete Random Variables			
5.02a 5.02b 5.02c	Probability distributions for general discrete random variables	a) Understand and be able to use discrete probability distributions. *Includes using and constructing probability distribution tables and functions relating to a given situation involving a discrete random variable.* *Any defined non-standard distribution will be finite.* b) Understand and be able to calculate the expectation and variance of a discrete random variable. *Includes knowing and being able to use the formulae* $\mu = \mathrm{E}(X) = \sum x_i p_i$ $\sigma^2 = \mathrm{Var}(X) = \sum (x_i - \mu)^2 p_i = \sum x_i^2 p_i - \mu^2$. [*Proof of these results is excluded.*] c) Know and be able to use the effects of linear coding on the mean and variance of a random variable.	
5.02d	The binomial distribution	d) Know and be able to use the formulae $\mu = np$ and $\sigma^2 = np(1-p)$ for a binomial distribution. *[Proof of these results is excluded.]* *For the underlying content on binomial distributions, see H240 sections 2.04b and 2.04c.*	
5.02e	The discrete uniform distribution	e) Know and be able to use the conditions under which a random variable will have a discrete uniform distribution, and be able to calculate probabilities and the mean and variance for a given discrete uniform distribution. *Includes use of the notation* $X \sim \mathrm{U}(n)$ *for the uniform distribution over the interval* $[1, n]$.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
5.02f 5.02g 5.02h	The geometric distribution	f) Know and be able to use the conditions under which a random variable will have a geometric distribution. *Includes use of the notation* $X \sim \mathrm{Geo}(p)$*, where* X *is the number of trials up to and including the first success.* g) Be able to calculate probabilities using the geometric distribution. *Learners may use the formulae* $\mathrm{P}(X=x)=(1-p)^{x-1}p$ *and* $\mathrm{P}(X>x)=(1-p)^{x}$*.* h) Know and be able to use the formulae $\mu=\frac{1}{p}$ and $\sigma^2=\frac{1-p}{p^2}$ for a geometric distribution. [*Proof of these results is excluded.*]	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
5.02i	The Poisson distribution	i) Understand informally the relevance of the Poisson distribution to the distribution of random events, and be able to use the Poisson distribution as a model. *Includes use of the notation $X \sim \mathrm{Po}(\lambda)$, where X is the number of events in a given interval.*	
5.02j		j) Understand and be able to use the formula . $\mathrm{P}(X = x) = e^{-\lambda}\dfrac{\lambda^x}{x!}$	
5.02k		k) Be able to calculate probabilities using the Poisson distribution, using appropriate calculator functions. *Learners are expected to have a calculator with the ability to access probabilities from the Poisson distribution.* [*Use of the Poisson distribution to calculate numerical approximations for a binomial distribution is excluded.*]	
5.02l		l) Know and be able to use the conditions under which a random variable will have a Poisson distribution. *Learners will be expected to identify which of the modelling conditions [assumptions] is/are relevant to a given scenario and to explain them in context.*	
5.02m		m) Be able to use the result that if $X \sim \mathrm{Po}(\lambda)$ then the mean and variance of X are each equal to λ.	
5.02n		n) Know and be able to use the result that the sum of independent Poisson variables has a Poisson distribution.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…